A-Level Physics Worksheet – Simple Harmonic Motion

Worksheet

Q1. Horizontal mass–spring system

A 0.30 kg block is attached to a horizontal spring on a frictionless table. The block is pulled to the right by 0.080 m and released from rest. It completes one oscillation every 0.90 s.

(a) Write the equation of motion $x(t)$, taking $x = 0$ at the equilibrium position and $+x$ to the right.

(b) Derive expressions for $v(t)$ and $a(t)$.

(c) Calculate $v_{\max}$ and $a_{\max}$.

(d) Determine the spring constant k.

(e) Find x, v and a at $t = 0.225\,\text{s}$.

Q2. Pendulum under small–angle approximation

A pendulum of length 0.60 m is given an initial horizontal velocity of $0.40\,\text{m}\,\text{s}^{-1}$ as it passes through its lowest point. Treat the motion as simple harmonic.

(a) Calculate the amplitude A of the resulting oscillations.

(b) Write $x(t)$ and $v(t)$, measuring x along the arc from the equilibrium position.

(c) Determine the first time the bob reaches a displacement of 0.020 m and its speed at that instant.

(d) Compute $a_{\max}$.

Q3. Graphical analysis I

Figure 1 shows the displacement–time graph for a mass on a spring.

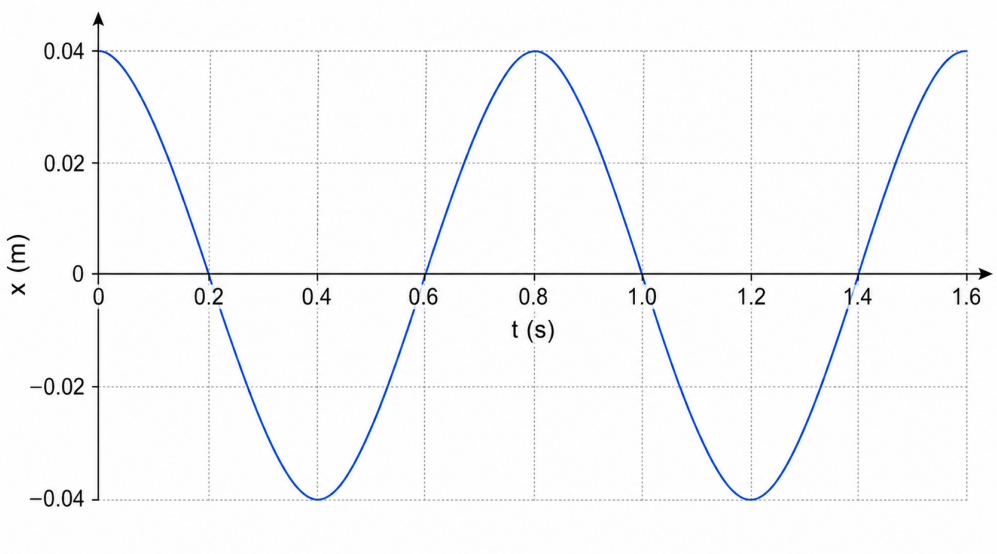


Figure 1: Displacement–time graph for Question 3

A 0.20 kg mass is attached to the spring.

(a) From the graph, determine A and T.

(b) Hence find ω and write $x(t)$.

(c) Calculate v and a at $t = 0.30\,\text{s}$.

(d) Determine the spring constant k.

Q4. Graphical analysis II

Figure 2 gives the velocity–time graph for a different SHM system.

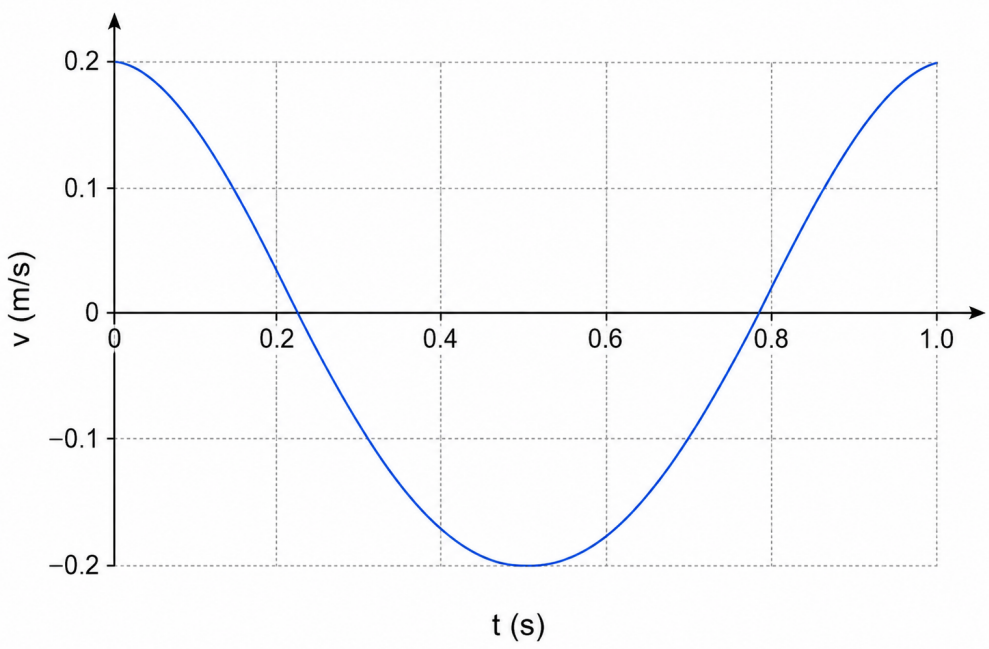


Figure 2: Velocity–time graph for Question 4

(a) Find the period and the velocity amplitude.

(b) Deduce the displacement amplitude A.

(c) Write expressions for $v(t)$ and $x(t)$.

(d) When $v = 0.15\,\mathrm{m\,s^{-1}}$ for the first time, calculate the displacement.

Q5. Vertical spring

A 0.20 kg mass hangs from a vertical spring. Pulling it down by 0.050 m from the equilibrium position and releasing it initiates SHM. Ten complete oscillations take 8.0 s.

(a) Find ω and k.

(b) Determine $v_{\max}$ and $a_{\max}$.

(c) Calculate the total mechanical energy of the oscillation.

Q6. SHM quantities at a given displacement

A 0.40 kg block oscillates on a horizontal spring of force constant $50\,\mathrm{N\,m^{-1}}$ with amplitude 0.060 m.

(a) Derive $v(x)$ and $a(x)$ for this system.

(b) Calculate the speed and acceleration when $x = 0.040\,\mathrm{m}$.

(c) How long after passing equilibrium does the block first reach $x = 0.040\,\mathrm{m}$?

Answer Key

Q1. (a) $x = 0.080\cos(6.98t)$

(b) $v = -0.558\sin(6.98t), \quad a = -3.90\cos(6.98t)$

(c) $v_{\max} = 0.558\ \mathrm{m\,s^{-1}}, \quad a_{\max} = 3.90\ \mathrm{m\,s^{-2}}$

(d) $k = m\omega^2 = 14.6\ \mathrm{N\,m^{-1}}$

(e) At $t = 0.225\,\mathrm{s}$: $x = 0,\ v = -0.558\,\mathrm{m\,s^{-1}},\ a = -3.90\,\mathrm{m\,s^{-2}}$

Q2. (a) $A = 0.099\,\mathrm{m}$

(b) $x = 0.099\sin(4.04t),\ v = 0.40\cos(4.04t)$

(c) $t = 0.050\,\mathrm{s},\ v = 0.392\,\mathrm{m\,s^{-1}}$

(d) $a_{\max} = 1.62\,\mathrm{m\,s^{-2}}$

Q3. (a) $A = 0.040\,\mathrm{m},\ T = 0.80\,\mathrm{s}$

(b) $\omega = 7.85\,\mathrm{rad\,s^{-1}},\ x = 0.040\cos(7.85t)$

(c) $x = -0.028\,\mathrm{m},\ v = -0.222\,\mathrm{m\,s^{-1}},\ a = 1.75\,\mathrm{m\,s^{-2}}$

(d) $k = 12.3\,\mathrm{N\,m^{-1}}$

Q4. (a) $T = 1.0\,\mathrm{s},\ v_{\mathrm{amp}} = 0.20\,\mathrm{m\,s^{-1}}$

(b) $A = 0.032\,\mathrm{m}$

(c) $v = 0.20\cos(6.28t),\ x = 0.032\sin(6.28t)$

(d) $x = 0.021\,\mathrm{m}$

Q5. (a) $\omega = 7.85\,\mathrm{rad\,s^{-1}},\ k = 12.3\,\mathrm{N\,m^{-1}}$

(b) $v_{\max} = 0.392\,\mathrm{m\,s^{-1}},\ a_{\max} = 3.09\,\mathrm{m\,s^{-2}}$

(c) $E = 0.015\,\mathrm{J}$

Q6. (a) $v(x) = \pm\omega\sqrt{A^2 - x^2},\ a(x) = -\omega^2 x$

(b) $v = 0.50\,\mathrm{m\,s^{-1}},\ a = -5.0\,\mathrm{m\,s^{-2}}$

(c) $t = 0.065\,\mathrm{s}$